# Introduction to Operations Research, Deterministic Models

## Lecture 13: Algorithms for Unconstrained Nonlinear Programming

Department of Industrial Engineering, Tsinghua University

Instructor: Junyi Liu

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# Recap: iterative descent methods for nonlinear programming

1. **Initialization:** start at some initial point $\mathbf{x}^0$;
2. **Iterative process:** iteratively generate a sequence of vectors $\mathbf{x}^1, \mathbf{x}^2, \ldots,$ according to

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r + \alpha^r \mathbf{d}^r}$$

- step size: $\alpha^r$
- direction: $\mathbf{d}^r$
- iterative solution: $\mathbf{x}^r$

3. **Descent property:** the step sizes $\{\alpha^r\}$ and directions $\{\mathbf{d}^r\}$ are chosen such that $f$ is decreased at each iteration, i.e.,

$$f(\mathbf{x}^{r+1}) = f(\mathbf{x}^r + \alpha^r \mathbf{d}^r) < f(\mathbf{x}^r)$$

hope to decrease $f$ all the way to the stationary solutions or (local) minimum.

# Gradient methods for unconstrained nonlinear programs

# Steepest descent method

**Idea:**

- By the first-order Taylor's expansion, with $\lambda$ sufficiently small and $\|\mathbf{d}\| = 1$,

$$f(\mathbf{x} + \lambda \mathbf{d}) \approx f(\mathbf{x}) + \lambda \nabla f(\mathbf{x})^\top \mathbf{d}$$

- $\nabla f(\mathbf{x})^\top \mathbf{d}$ is the slope of the function value gap $f(\mathbf{x} + \lambda \mathbf{d})$ at $\lambda = 0$ along the direction $\mathbf{d}$
- minimize $\left\{\nabla f(\mathbf{x})^\top \mathbf{d} : \|\mathbf{d}\| = 1\right\}$, the optimal solution $\mathbf{d}^* = -\dfrac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$ (given $\nabla f(\mathbf{x}) \neq 0$)
- Given $\nabla f(\mathbf{x}) \neq 0$ and $\lambda$ is sufficiently small,

$$f(\mathbf{x} + \lambda \mathbf{d}^*) - f(\mathbf{x}) = -\lambda \|\nabla f(\mathbf{x})\| + o(\lambda) < 0$$

**Steepest descent method**

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r - \alpha^r \, \nabla f(\mathbf{x}^r)}$$

is one of the gradient methods.

# Gradient methods: descent directions

More generally, we are interested to find a vector $\mathbf{d}$ such that $f(\mathbf{x}+\lambda\mathbf{d}) < f(\mathbf{x})$ with sufficiently small $\lambda$.

## Definition (descent direction)

We say $\mathbf{d}$ is a descent direction of $f$ at $\bar{\mathbf{x}}$ if $f(\bar{\mathbf{x}}+\lambda\mathbf{d}) < f(\bar{\mathbf{x}})$, for any $\lambda \in (0,\delta)$, with some $\delta > 0$

Any vector $\mathbf{d} \in \mathbb{R}^n$ satisfying $\nabla f(\bar{\mathbf{x}})^\top \mathbf{d} < 0$ must be a descent direction. Why?

This is because: by the first-order Taylor's expansion, for any $\mathbf{d}$ satisfying $\nabla f(\bar{\mathbf{x}})^\top \mathbf{d} < 0$,

$$f(\bar{\mathbf{x}}+\lambda\mathbf{d}) - f(\bar{\mathbf{x}}) = \lambda \nabla f(\bar{\mathbf{x}})^\top \mathbf{d} + o(\lambda) < 0 \quad \text{for sufficiently small } \lambda > 0.$$

# Gradient methods

From a point $\mathbf{x}$ with $\nabla f(\mathbf{x}) \neq 0$, we can move towards a direction $\mathbf{d}$ with $\nabla f(\mathbf{x})^\top \mathbf{d} < 0$ to decrease the function value with the hope to get closer to the minimzer.

**Gradient methods:**

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r + \alpha^r \mathbf{d}^r \text{ for } r = 0, 1, \ldots,}$$

where $\nabla f(\mathbf{x}^r)^\top \mathbf{d}^r < 0$; and $\alpha^r$ is carefully chosen so that it is indeed a descent algorithm.

**General principles:**

- there are various choices of the direction $\mathbf{d}^r$ and the stepsize $\alpha^r$ in a gradient method;
- there is no single gradient method that can be recommended for all or even most problems;
- we need to understand the performance of different gradient methods and identify the practical contexts in which the usage of particular gradient methods are appropriate.

# Choices of directions in gradient methods

The directions in many gradient methods have the form

$$\boxed{\mathbf{d}^r = -D^r \nabla f(\mathbf{x}^r)}$$

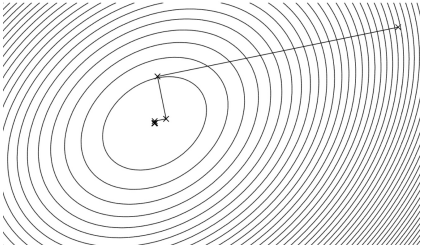
with a symmetric PD matrix $D^r$.

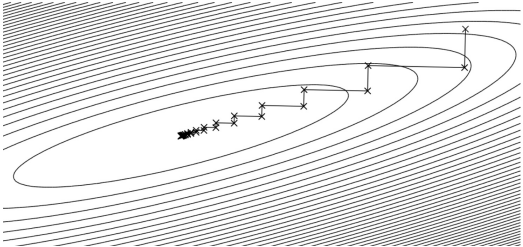
This is indeed a descent direction as

$$(\mathbf{d}^r)^\top \nabla f(\mathbf{x}^r) = -\nabla f(\mathbf{x}^r)^\top D^r \nabla f(\mathbf{x}^r) < 0, \text{ when } \nabla f(\mathbf{x}^r) \neq 0$$

# Steepest descent method: $D^r = I$

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r - \alpha^r \nabla f(\mathbf{x}^r)}$$





Advantage: easy and fast to compute the iterative points

Disadvantage: zig-zag behavior, slow convergence when the surface of $f$ is very "elongated"

# Newton's method: $D^r = (\nabla^2 f(\mathbf{x}^r))^{-1}$

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r - \alpha^r \left(\nabla^2 f(\mathbf{x}^r)\right)^{-1} \nabla f(\mathbf{x}^r)}$$

**Idea:**

- By the second-order expansion, the quadratic approximation around the current point $\mathbf{x}^r$ is

$$\hat{f}(\mathbf{x}^r + \mathbf{d}; \mathbf{x}^r) \triangleq f(\mathbf{x}^r) + \nabla f(\mathbf{x}^r)^\top \mathbf{d} + \frac{1}{2}\, \mathbf{d}^\top \nabla^2 f(\mathbf{x}^r) \mathbf{d}$$

Suppose that $\nabla^2 f(\mathbf{x}^r)$ is a PSD matrix. To minimize $\hat{f}(\mathbf{x}^r + \mathbf{d}; \mathbf{x}^r)$ over $\mathbf{d} \in \mathbb{R}^n$, the necessary and sufficient optimality condition is

$$\nabla^2 f(\mathbf{x}^r)\, \mathbf{d}^r + \nabla f(\mathbf{x}^r) = 0 \quad \Longrightarrow \quad \mathbf{d}^r = -(\nabla^2 f(\mathbf{x}^r))^{-1} \nabla f(\mathbf{x}^r)$$

- $\mathbf{d}^r$ is a descent direction when $D^r = (\nabla^2 f(\mathbf{x}^r))^{-1}$ is PD

Advantage: converge faster, especially in the local region of the optimal solutions;

Disadvantage: $\nabla^2 f(\mathbf{x}^r)$ may not be PD; computation of $(\nabla^2 f(\mathbf{x}^r))^{-1}$ is expensive

# Choice of step sizes

1 **constant step size:** $\alpha^r = \alpha$, for all $r = 0, 1, \ldots,$

- simplest rule to implement, but may not converge if $\alpha$ is too large;
- may be too slow to converge if $\alpha$ is too small.

2. **diminishing step size:** $\lim_{r\to\infty} \alpha^r = 0.$ and $\sum_{r=1}^{\infty} \alpha^r = \infty$

- descent property only holds when step size becomes sufficiently small;
- $\sum \alpha^r = \infty$ is imposed so that progress does not become too slow;
- good theoretical guarantees; unless the right sequence is chosen, can also be a slow method.

3. **exact step size:** $\alpha^r \in \text{argmin}_{\alpha\geq 0} f(\mathbf{x}^r + \alpha \mathbf{d}^r)$

- need to solve a minimization problem;
- no need to be very precise at each step.

# Extension: convergence of steepest descent method

For a gradient method,

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r + \alpha^r \mathbf{d}^r \text{ for } r = 0, 1, \ldots,},$$

## Theorem (diminishing step size)

Assume

1. $f$ is a continuously differentiable function bounded from below with Lipschitz gradient: $\exists L > 0$, such that $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
2. the descent directions $\mathbf{d}^r = -\nabla f(\mathbf{x}^r)$ (could be more general)
3. the diminishing step size satisfies $\lim_{r\to\infty} \alpha^r = 0$, and $\sum_r \alpha^r = \infty$

Then, every limit point of the solution sequence $\{\mathbf{x}^r\}$ if exists is a stationary point, i.e.,

$$\text{if } \lim_{r(\in\mathcal{K})\to\infty} \mathbf{x}^r = \bar{\mathbf{x}}, \text{ then } \nabla f(\bar{\mathbf{x}}) = 0$$

# Non-differentiable convex programs

Suppose $f(\bullet) : \mathbb{R}^n \to \mathbb{R}$ is a convex function

$$\text{minimize} \quad f(\mathbf{x})$$

By the convexity, $f(\mathbf{x}) \geq f(\mathbf{x}^r) + (\mathbf{g}^r)^\top (\mathbf{x} - \mathbf{x}^r)$, for any $\mathbf{g}^r \in \partial f(\mathbf{x}^r)$

**Subgradient method**

$$\boxed{\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha^r \mathbf{g}^r, \text{ with } \mathbf{g}^r \in \partial f(\mathbf{x}^r)}$$

The negative of a subgradient may not be a descent direction.

It necessitates a careful choice of step sizes $\{\alpha^r\}$ to ensure convergence.

We keep track of the best iterate point $\mathbf{x}^k_{\text{best}}$ among $k$ iterations with $f(\mathbf{x}^k_{\text{best}}) = \min_{r=0,\ldots,k} f(\mathbf{x}^r)$.

# The role of convergence analysis

Merits of the convergence analysis include:

1. **delineate the range of applicability of various methods and appropriate parameter choices:**
   e.g., selection of step sizes, the cost function is once or twice differentiable, convex or nonconvex, Lipschitz gradient, or strongly convex.
2. **help us understand the qualitative and quantitative behavior of various methods:**
   e.g., the role of initial points, whether the sequence $\{\mathbf{x}_k\}$ or $\{f(\mathbf{x}_k)\}$ converges, what is the limit point of $\{\mathbf{x}_k\}$, how fast does the algorithm converge, how does the problem structure affect the convergence rate?
3. **provide the guidelines for choosing a few algorithms** for further experimentation out of numerous candidate algorithms

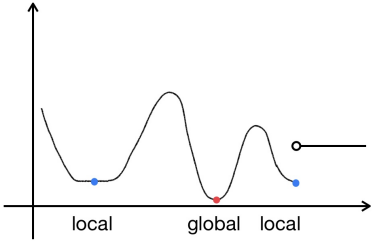
Limitations of the convergence analysis include:

- assumptions may be hard to be verified or too strong;
- the convergence rate is the worst result, it may perform faster in practice for special types of problems;
- mathematical analysis is too dry to be understood.

# Constrained nonlinear programs

# Constrained nonlinear programs

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $S$ be a nonempty set in $\mathbb{R}^n$. Consider the minimization of $f(\mathbf{x})$ over $S$. We say $\bar{\mathbf{x}} \in S$ is a




- **global minimizer:** if $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$;
- **local minimizer:** $\exists\, N_\epsilon(\bar{\mathbf{x}})$ such that $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for any $\mathbf{x} \in N_\epsilon(\bar{\mathbf{x}}) \cap S$;

# Geometric optimality conditions

**Recall:**

- we say $\mathbf{d}$ is a descent direction of $f$ at $\bar{\mathbf{x}}$ if $f(\bar{\mathbf{x}} + \lambda \mathbf{d}) < f(\bar{\mathbf{x}})$, $\forall \lambda \in (0, \delta)$, with some $\delta > 0$
- $F_0(\bar{\mathbf{x}}) \triangleq \{\mathbf{d} \in \mathbb{R}^n : \nabla f(\bar{\mathbf{x}})^\top \mathbf{d} < 0\} \subseteq F(\bar{\mathbf{x}})$ belongs to the set of descent directions

For unconstrained nonlinear programs, if a point $\bar{\mathbf{x}}$ is a local optimum, then the function value does not decrease along any direction at $\bar{\mathbf{x}}$.

$$\boxed{\bar{\mathbf{x}} \text{ is a local minimum}} \longrightarrow \boxed{F_0(\bar{\mathbf{x}}) = \emptyset} \longleftrightarrow \boxed{\nabla f(\bar{\mathbf{x}}) = 0}.$$

For constrained nonlinear programs, if a point $\bar{\mathbf{x}} \in S$ is a local optimum, then the function value does not decrease along any feasible direction at $\bar{\mathbf{x}}$.

$$\boxed{\bar{\mathbf{x}} \text{ is a local minimum}} \longrightarrow \boxed{F_0(\bar{\mathbf{x}}) \cap \text{ feasible direction set} = \emptyset}.$$

# Linearized Cone

| | | | |
|---|---|---|---|
| Minimize | $f(\mathbf{x})$ | | (objective function) |
| subject to | $g_i(\mathbf{x}) \leq 0,$ | for $i \in \mathcal{I}$ | (inequality constraints) |
| | $g_i(\mathbf{x}) = 0,$ | for $i \in \mathcal{E}$ | (equality constraints) |

The constrained set $S \triangleq \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0, \forall i \in \mathcal{I};\ g_i(\mathbf{x}) = 0, \forall i \in \mathcal{E}\} = \bigcap_{i \in \mathcal{I} \cup \mathcal{E}} S_i$ where $S_i \triangleq \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0\}$ for $i \in \mathcal{I}$, and $S_i \triangleq \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) = 0\}$ for $i \in \mathcal{E}$.

Given a point $\bar{\mathbf{x}} \in S$, what would be a feasible direction?

1. for $i \in \mathcal{I}$ with $g_i(\bar{\mathbf{x}}) < 0$: any $\mathbf{d} \in \mathbb{R}^n$ is a feasible direction of $S_i$;
2. for $i \in \mathcal{I}$ with $g_i(\bar{\mathbf{x}}) = 0$: any $\mathbf{d} \in \mathbb{R}^n$ satisfying $\nabla g_i(\bar{\mathbf{x}})^\top \mathbf{d} < 0$ is a feasible direction of $S_i$;
3. for $i \in \mathcal{E}$: the function value is likely to remain the same towards the direction $\mathbf{d} \in \mathbb{R}^n$ satisfying $\nabla g_i(\bar{\mathbf{x}})^\top \mathbf{d} = 0$.

**The feasible direction should only be affected by the constraint functions with indexes in the latter two cases.** Define the active index set $\mathcal{A}(\bar{\mathbf{x}}) \triangleq \mathcal{E} \cup \{i : g_i(\bar{\mathbf{x}}) = 0, i \in \mathcal{I}\}$.

# Linearized Cone

## Definition

The linearized cone at $\bar{\mathbf{x}} \in S$ is defined as

$$L_S(\bar{\mathbf{x}}) \triangleq \left\{ \mathbf{d} : \nabla g_i(\bar{\mathbf{x}})^\top \mathbf{d} \leq 0, \forall\, i \in \mathcal{A}(\bar{\mathbf{x}}) \cap \mathcal{I}; \nabla g_i(\bar{\mathbf{x}})^\top \mathbf{d} = 0, \forall\, i \in \mathcal{E} \right\}.$$

**Interpretation:** for the vector in $L_S(\bar{\mathbf{x}})$,

1. it is perpendicular to $\{\nabla g_i(\bar{\mathbf{x}})\}_{i \in \mathcal{E}}$ such that the the function value $g_i(\mathbf{x})$ is likely to remain the same towards that direction;
2. its angle with $\{\nabla g_i(\bar{\mathbf{x}})\}_{i \in \mathcal{A}(\bar{\mathbf{x}}) \cap \mathcal{I}}$ is greater or equal to 90°, so that the function value $g_i(\mathbf{x})$ is likely to decrease towards that direction;
3. is not affected by the inactive constraints in $\mathcal{I}$.

**Properties:**

1. $L_S(\bar{\mathbf{x}})$ is a polyhedral set;
2. Different algebraic representations of $S$ may result in much different linearized cones.

eg: $S_1 = \{x : x \geq 3\}$, $S_2 = \{x : (3-x)^3 \leq 0\}$. $L_{S_1}(3) = \{d : d \geq 0\}$, $L_{S_2}(3) = \mathbb{R}$.

# KKT necessary conditions

## Karush-Kuhn-Tucker (KKT) necessary conditions

If $\bar{\mathbf{x}} \in S$ is a local optimum of the problem and $\{\nabla g_i(\bar{\mathbf{x}})\}_{i \in \mathcal{A}(\bar{\mathbf{x}})}$ are linearly independent, then the following three equivalent conditions hold:

1. $F_0(\bar{\mathbf{x}}) \cap L_S(\bar{\mathbf{x}}) = \emptyset$;
2. there exist scalars $\{u_i\}_{i \in \mathcal{I} \cap \mathcal{A}(\bar{\mathbf{x}})} \subset \mathbb{R}_+$ and $\{v_i\}_{i \in \mathcal{E}} \subset \mathbb{R}$ such that

$$\nabla f(\bar{\mathbf{x}}) + \sum_{i \in \mathcal{I} \cap \mathcal{A}(\bar{x})} u_i \nabla g_i(\bar{\mathbf{x}}) + \sum_{i \in \mathcal{E}} v_i \nabla g_i(\bar{\mathbf{x}}) = 0$$

3. there exist scalars $\{u_i\}_{i \in \mathcal{I}}$ and $\{v_i\}_{i \in \mathcal{E}}$ such that:

$$\begin{aligned}
\nabla f(\bar{\mathbf{x}}) + \sum_{i \in \mathcal{I}} u_i \nabla g_i(\bar{\mathbf{x}}) + \sum_{i \in \mathcal{E}} v_i \nabla g_i(\bar{\mathbf{x}}) = 0 & \qquad \text{(stability condition)} \\
u_i g_i(\bar{\mathbf{x}}) = 0, \ \forall i \in \mathcal{I} & \qquad \text{(complementary slackness conditions)} \\
u_i \geq 0, \ \forall i \in \mathcal{I} & \qquad \text{(dual feasible conditions)} \\
g_i(\bar{\mathbf{x}}) \leq 0, \ \forall i \in \mathcal{I}, \qquad g_i(\bar{\mathbf{x}}) = 0, \forall i \in \mathcal{E} & \qquad \text{(primal feasible conditions)}
\end{aligned}$$

The point $\bar{\mathbf{x}} \in S$ satisfying the above conditions is called a KKT point.

# KKT sufficient conditions

Consider the problem

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & \mathbf{x} \in S \triangleq \{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0,\ \forall i \in \mathcal{I}; \quad g_i(\mathbf{x}) = 0,\ \forall i \in \mathcal{E}\} \end{aligned}$$

## Theorem

Suppose that $f(\bullet)$ and $\{g_i(\bullet)\}_{i\in\mathcal{I}}$ are differentiable convex functions, and $\{g_i(\bullet)\}_{i\in\mathcal{E}}$ are linear functions. If $\bar{\mathbf{x}} \in S$ is a KKT point, i.e., there exists scalars $\{u_i\}_{i\in\mathcal{A}(\bar{\mathbf{x}})\cap\mathcal{I}} \subset \mathbb{R}_+$ and $\{v_i\} \subset \mathbb{R}$ satisfying

$$\nabla f(\bar{\mathbf{x}}) + \sum_{i\in\mathcal{A}(\bar{\mathbf{x}})\cap\mathcal{I}} u_i \nabla g_i(\bar{\mathbf{x}}) + \sum_{i\in\mathcal{E}} v_i \nabla g_i(\bar{\mathbf{x}}) = 0,$$

then $\bar{\mathbf{x}}$ is a global optimal solution of the convex problem.

# Computation of KKT points

Consider

$$\begin{array}{ll} \text{minimize} & (x_1-1)^2+x_2^2 \\ \text{subject to} & g_1(x_1,x_2)=x_1-x_2^2\le 0 \end{array}$$

**Observations:**

(1) it is a nonconvex program with one single inequality constraint;
(2) $\nabla g_1(x_1,x_2)=(1,-2x_2)^\top \neq (0,0)^\top$
(3) $(\bar{x}_1,\bar{x}_2)$ is a KKT point if and only if: $\exists\, u$ such that

$$\begin{cases} \begin{pmatrix} 2(\bar{x}_1-1) \\ 2\bar{x}_2 \end{pmatrix} + u \begin{pmatrix} 1 \\ -2\bar{x}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ u(\bar{x}_1-\bar{x}_2^2)=0 \\ u\ge 0,\ \bar{x}_1-\bar{x}_2^2\le 0 \end{cases}$$

(4) if $u=0$, then $(x_1,x_2)=(1,0)$ which is infeasible. Thus we must have $u>0$.
(5) $(1/2,1/\sqrt{2})$ and $(0,0)$ are two KKT points, which are the candidates for local minimum.

# Lagrangian Dual Problem

# Lagrangian Dual Problem

**Primal Problem**

$$
\begin{array}{rll}
p^* = & \text{minimize} & f(\mathbf{x}) \\
 & \text{subject to} & g_i(\mathbf{x}) \leq 0 \quad \text{for } i = 1, \ldots, m \\
 & & h_i(\mathbf{x}) = 0 \quad \text{for } i = 1, \ldots, \ell \\
 & & \mathbf{x} \in X \subseteq \mathbb{R}^n
\end{array}
$$

**Lagrangian function:** $L(\bullet, \bullet, \bullet) : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^\ell \to \mathbb{R}$ with

$$L(\mathbf{x}, \mathbf{u}, \mathbf{v}) = f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x})$$

**Lagrangian dual function:** $\theta(\bullet, \bullet) : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}$ with

$$\theta(\mathbf{u}, \mathbf{v}) = \inf \left\{ f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X \right\}$$

**Lagrangian Dual Problem**

$$
\begin{array}{rll}
d^* = & \text{maximize} & \theta(\mathbf{u}, \mathbf{v}) \\
 & \text{subject to} & \mathbf{u} \geq 0
\end{array}
$$

# Construction of the Dual Problem

1. Identify the inequality and equality constraints to be penalized, and the "ground-set" $X$
2. Create the Lagrangian function $L(\mathbf{x}, \mathbf{u}, \mathbf{v}) = f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x})$
3. Create the Lagrangian dual function $\theta(\mathbf{u}, \mathbf{v}) = \inf \left\{ f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X \right\}$
4. Construct the dual problem maximize$\{\, \theta(\mathbf{u}, \mathbf{v}) : \mathbf{u} \geq 0 \}$

# Dual Problem of Linear Programming

## Primal LP

$$\begin{array}{ll} \text{minimize} & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \geq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$$

**Identification:** $X \triangleq \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq 0\}$ and $\mathbf{g}(\mathbf{x}) = \mathbf{b} - \mathbf{A}\mathbf{x}$

**Lagrangian function:** $L(\mathbf{x}, \mathbf{u}) = \mathbf{c}^\top \mathbf{x} + \mathbf{u}^\top (\mathbf{b} - \mathbf{A}\mathbf{x})$

**Lagrangian dual function:** $\theta(\mathbf{u}) = \min_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{u})$

## Dual problem?

# Dual Problem of Convex Quadratic Programming

## Primal QP

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}\mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \geq \mathbf{b} \end{array}$$

where $\mathbf{Q}$ is a symmetric and positive definite matrix.

**Identification:** $X = \mathbb{R}^n$ and $\mathbf{g}(\mathbf{x}) = \mathbf{b} - \mathbf{A}\mathbf{x}$

**Lagrangian function:** $L(\mathbf{x}, \mathbf{u}) = \frac{1}{2}\mathbf{x}^\top \mathbf{Q}\mathbf{x} + \mathbf{c}^\top \mathbf{x} + \mathbf{u}^\top (\mathbf{b} - \mathbf{A}\mathbf{x})$

**Lagrangian dual function:** $\theta(\mathbf{u}) = \min_{\mathbf{x} \in \mathbb{R}^n} L(\mathbf{x}, \mathbf{u})$

## Dual problem?

# Reading materials

Section 7.5, 7.6, 7.8 *Operations Research: mathematical programming* by Wayne L. Winston